BT301 CURRENT PAPERS SOLUTIONS

Q. Difference between genetics and genomics? 2.

Genetics • Genetics is the study of heredity, or how the characteristics of living organisms are transmitted from one generation to the next generation through DNA

Genomics, in contrast, is the study of the entirety of an organism's genes – called the genome

Q. What is size diff between LINEs and SINEs?

SINEs (short interspersed elements) 200-300 bp • 100,000 copies

• LINEs (long interspersed elements) 1-5 kb • 10-10,000 copies

Q. What is green biotechnology and give an example?

Green Biotechnology It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones.

Example Bt corn from Syngenta Bt cotton from Monsanto

Q. What is Novel Diagnostic?

- Complex systems biology, developmental genetics.
- To provide platform for microchips and DNA microarrays.
- Gene expression - RNA
- Complex systems biology, developmental genetics and genomics

Q. Write only name of 4 Major technique used in genome characterization

1) Cloning 2) Hybridization 3) PCR Amplification 4) Sequencing 5) Computational tool

Q. What is Insertional Inactivation?

Insertional inactivation is the inactivation of a gene upon insertion of another gene inside in its place or within its coding sequence

Q. How many Standard Amino Acids are there? Write name of any 5?

There are 20 amino acids know as standard amino acids.

1) Leucine 2) Isoleucine 3) Valine 4) Proline 5) Methionine

Q. What are various factors which affect mobility of DNA in the Gel?

- Concentration of gel either high concentration or lower concentration.
- Ionic strength of buffer.
- Voltage
- Ethidium bromide (A chemical used as a stain to dye gel)

Q. What is the role of Electroporation?

In Electroporation electric field is applied to cells in order to increase permeability of the cell membrane to take DNA

Q. Define Autoradiography
A technique using X-ray film to visualize fragments of molecules that have been radioactively labeled Eg. It can be used to analyze the length and number of DNA fragments separated by gel electrophoresis

Q. Write main sources and usage of citric acid /2
Main source of citric acid is citrus fruits but in industrial production citric acid is obtained by Aspergillus Niger. It is used to preserve food and to flavoring food like vinegar

Q. Define Gene
A **gene** is the basic physical and functional unit of heredity. **Genes**, which are made up of DNA, act as instructions to make molecules called proteins

Q. Define Isopynic Centrifugation
A method where the components of a sample (e.g. DNA) are separated on the basis of their density.

Q= Define Autoradiograpy?(2)

A technique using X-ray film to visualize fragments of molecules that have been radioactively labeled. For example, it can be used to analyzed the length and number of DNA fragments separated by gel electrophoresis.

Q=Define :Euchromatic & hetrchromatin? (2)

Highly condensed chromatin is known as **heterochromatin**. The more extended form is known as **euchromatin.**

Q=what is the main components consisting the RNA molecule.?

- Each ribonucleotide is composed of three components:
  - A ribose sugar
  - A Nitrogenous Base
  - A Phosphoric acid

Q. What is .palindromic sequence?

A DNA locus whose 5'-to-3' sequence is identical on each DNA strand. The sequence is the same when one strand is read left to right and the other strand is read right to left. Recognition sites of many restriction enzymes are palindromic.

Q. Define the function of Plasmid DNA

- Plasmids are mostly circular DNA molecules that replicate separately from the host chromosome
- Plasmid help bacteria to survive stress

Q. Ancient Pregnancy Test

A papyrus described a test in which a woman who might be pregnant could urinate on wheat and barley seeds over the course of several days: "If the barley grows, it means a male child. If the wheat grows, it means a female child. If both do not grow, she will not bear at all."

Q. what is pseudo gene?

Pseudogenes are **genes** that have lost their function. They have lost their **gene** expression in the cell or their ability to code protein. The term was coined in 1977. Pseudogenes can result from mutations in a **gene** whose product is not needed for the survival of the organism

Q= Differentiate between multimer,ologimer and protomer?

• A multi-subunit protein is also referred to as a **multimer**. A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer. Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.

Q= write one differential point between super coiled and relaxed loops? long question?

Genetic Material in Bacteria some of the loops are super-coiled while some are relaxed. Relaxed loops are formed as a result of a nick introduced into super-coiled loops by a cellular DNase. If a super-coiled DNA molecule receives a nick, the strain of under winding is immediately removed, and all the super-coiling is lost.

The super-coiled loops are dynamic structures which change during cell growth & division. An E. coli chromosome is estimated to have about 400 supercoiled loops. The DNA compaction in a bacterial cell is contributed by supercoiling of loops, macromolecular crowding and DNA binding proteins.

Q: What is Selfish DNA? (2)

Selfish DNA is a term for sequences of **DNA** that have two distinct properties: the **DNA** sequence spreads by forming additional copies of itself within the genome; and. it makes no specific contribution to the reproductive success of its host organism. (It may or may not have significant negative effects.)

Q=Side effect of ethidium bromide?

Ethidium bromide can bind with DNA, it is highly toxic as a mutagen. It may potentially cause carcinogenic or teratogenic effects, although no scientific evidence showing either health effect has been found. Exposure routes of ethidium bromide are inhalation, ingestion, and skin absorption. An acute exposure to ethidium bromide causes irritation of the mouth, upper respiratory tract, skin, and eyes.

Q: Importance of origin region in plasmids?

The replicon is comprised of the origin of replication (ORI) and all of its control elements. The ORI is the place where DNA replication begins, enabling a plasmid to reproduce itself as it must to survive within cells.

- When plasmids are present in microbes, whether they have narrow host range or wide host range.
- Plasmids encode only a few of the proteins required for their own replication
- Host range of plasmid is determined by *ori* region (Point where DNA replication mainly starts)
- Plasmids whose *ori* derived from Col E1 have restricted host range
- RP4 and RSF1010 and many plasmids from *Staphylococcus aureus* have broad host range

Q. Write to most significant function of Nucleic Acid / DNA Function? /2

The main **functions** is store and transfer genetic information. To use the genetic information to direct the synthesis of new protein. The deoxyribo**nucleic acid** is the storage for place for genetic information in the cell

Q. Define Natural Plasmid? /2

Plasmid which were not constructed in vitro for the sole purpose of cloning are called natural plasmids Example Col E1 is natural occurring plasmid.

Q. Write 2 main factors that result in High yield of Plasmid DNA /3

2 main factors that result in High yield of plasmid are

1) Concentration of chemical that is used Calcium chloride
2) Temperature

Q. Write line for Influenza. /2

The influenza virion (as the infectious particle is called) is roughly spherical. It is an enveloped virus – that is, the outer layer is a lipid membrane which is taken from the host cell in which the virus multiplies

Q. Explain Technique of Electroporation

Electroporation

- A rapid and simple technique for introducing cloned genes into a wide variety of microbial, plant and animal cell, including *E. coli,* is electroporation
- In electroporation, electric field is applied to cells in order to increase permeability of the cell membrane to take DNA

Transformation with other organisms

- *E. coli* often remains the host of choice for cloning
- *Bacillus subtilis (preferable choice because it is natural competent)* is a particularly attractive alternative cloning host
- Animal cells, protoplasts (without cell wall) of yeast and plant are susceptible to transformation by liposomes

JOIN VU BIO ANIMALS

- Other artificial method for transformation include gene gun
- Lipofaction is a technique in which we use liposomes to transform protoplasts. Liposomes are vesicles made up of phospholipids and are positively charged. They are positively charged because mainly they are made from cationic lipids. They have the ability to enclose negatively charged DNA. After encapsulation liposomes fuse with the cytoplasmic membrane and then it can deliver DNA from the vesicle to the target cell. So lipofaction is also used to transform different type of host cells.

Q. Write Nomenclature of Restriction Enzyme Modification

Nomenclature

After the discovery of large number of Type II endonucleases, scientists proposed that there must be a naming mechanism which we can call as the nomenclature of the restriction endonucleases.

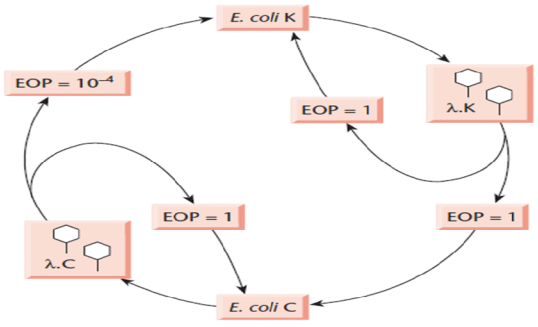
- A suitable system was proposed by Smith and Nathans (1973)
- The species name of host organisms is identified by the first letter of genus and first two letters of specific epithet-
- *E. coli* = Eco
- *H. influenzae* = Hin
- Strain identification is written as *Eco*K
- In case, host strain has several restriction and modification systems, these are identified by roman numerals, for example, in case of *H. influenzae*
- *HindI, HindII, HindIII* etc
- All restriction enzymes have general name endonuclease R and modification-methylase M followed by the system name, for example, in case of *H. influenzae*
- R. *HindIII* or M. *HindIII*

Q. Explain R-M Phenomena

- During 1960s phage biologist elucidated the phenomenon of restriction and modification

Host-controlled restriction/ modification

- Restriction system allow bacteria to monitor the origin of incoming DNA
- When the incoming DNA is a bacteriophage genome, the effect is to reduce the efficiency of plating (Efficiency of plating means the number of plaques that are produced during the lysis) i.e. to reduce the number of plaques in plating test
- The phenomenon of restriction and modification were well illustrated and studied by the behavior of phage λ on two *E. coli* strains
- *E. coli* C and *E. coli* K
- Restrictive host must, of course protects its own DNA from restriction endonucleases by modification
- Modification involves methylation of certain bases that constitute recognition sites



JOIN VU BIO ANIMALS

Q. Write Benefits of Genomic Research in Molecular Medicine

Genomes Research – Molecular Medicine

- Improve diagnosis of disease
- Detect genetic predispositions to disease (cancer, diabetes etc.)
- Create drugs based on molecular information
- Use gene therapy and control systems as drugs

Q. Chargaff Rule

The work of Chargaff led him to following conclusions, also called "Chargaff Rules":-

1. Base composition of DNA varies from one species to another. 2. The DNA isolated from different tissues of the same species has the same base composition. 3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment. 4. In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C). 5. It means that the sum of the purine residues equals the sum of the pyrimidine residues (AG=TC).

Q= Biotechnology define and write its branches ?

Biotechnology is the use of microbes, animal/plant cells and their products to synthesize, break down or transform materials. Primarily it includes the use of recombinant DNA technology and genetic engineering techniques to improve upon the quality of processes.

1) **White Biotechnology:** Development of processes and microorganisms for Industrial processes. e.g; Enzyme production
2) **Red Biotechnology:** It is concerned with the discovery and development of innovative drugs and treatments.
3) **Green Biotechnology:** It is concerned with the modification of the genetic composition of plants to enhance existing traits or add new ones. An additional gene was inserted into the normal genome of this corn that triggers the production of proteins that then protect the plant from harmful insects. This reduces the need for pesticides and increases crop yields. **Example:** Bt corn from Syngenta Bt cotton from Monsanto

Q. Work of Franklin and Wlkins

Work of Franklin & Wilkins (1950s):

- Rosalind Franklin and Maurice Wilkins performed the x-ray diffraction analysis of DNA fibres.
- They showed that DNA produces a characteristic x-ray diffraction pattern.
- From this pattern, they made two important findings:-
- DNA molecules are helical.
- The helices have two periodicities along their long axis, a primary one of 3.4 Å and a secondary one of 34 Å.

Q. Explain Structure of proteins

Primary structure of proteins:

Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain. Each type of protein has a unique amino acid sequence. Proteins are different from one another on the basis of their amino acid sequence.

Secondary structure of proteins:

Secondary structure of proteins refers to the local conformation of some part of a polypeptide. A few types of secondary structures are particularly stable and occur widely in proteins.

The most prominent are:- 1. α-helix:- 2. β- conformations:-

Tertiary Structure

The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's tertiary structure. It includes longer-range aspects of amino acid sequence. Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein. Large polypeptide chains usually fold into two or more globular clusters known as **domains,** which often give these proteins a bi- or multilobal appearance.

Quaternary Structure of Proteins:

- Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different. The spatial arrangement of these subunits is known as a protein's quaternary structure.

JOIN VU BIO ANIMALS